

Alejandro Alfaro

Senior Firmware Engineer with 5+ years building IEC 62304-compliant embedded systems for medical wearables — Zephyr RTOS, BLE, and bootloader/OTA architecture from prototype to production.

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Skills & Education

Firmware Experiences	Zephyr Modular Architecture, Bootloader Design , BLE and IoT
Tech Proficiencies	Modern C Design Patterns, IPC & Concurrency , Build/Test Pipelines
Domain Knowledge	Medical IoT Wearables Development, IEC 62304 , CE Mark/FDA 510(k) submissions
Higher Education	Bachelor's and Master's in Mechanical Engineering , Northwestern Class of 2020

Also: RTOS, C, C++, I2C, SPI, UART, PDM, ADC, Git, CMake, ARM GCC/Clang, GDB, JTAG/SWD, Logic Analyzer

Relevant Experience

Senior Firmware Engineer

Remote from San Francisco, CA

SIBEL HEALTH

January 2025 – Present

- Led development of two wearable medical devices — an ICU-grade multi-lead ECG sensor for hospital use and an acousto-mechanical sensor for therapeutics and clinical research — securing over \$10M in grant funding.
- Architected embedded systems for both products from component selection and EVT prototype procurement through early technical de-risking of hardware and firmware designs.
- Initiated and owned transition to firmware services/modules-oriented design using ZBus, Zephyr's pub-sub channel based IPC framework, positively impacting all our products by enabling re-use of code up to 50% in some cases and reducing overall development time/cost.
- Designed a 3-image bootloader strategy (bootloader + app + firmware loader) that doubled available ROM from 400KB to 800KB on memory-constrained platforms, added OTA recovery mode, and isolated BLE image downloads from the application for improved security.
- Driving a huge refactor of a legacy fragile platform built by offshore consultants with only one developer (me) and AI agents following IEC 62304 practices and using the latest tools/harnesses.

TECHNICAL PROJECT MANAGER

September 2023 – January 2025

- Managed the Design History File (DHF) for two next-gen platforms through Phase 1–3 of a Medical Device PDP, collaborating with multi-disciplinary teams to finalize design inputs before production.
- Reduced engineering costs by ~\$1–2M by outsourcing hardware and firmware development to an offshore partner with strong silicon vendor relationships, accelerating component procurement.
- Coordinated between the offshore team and Sibel leadership through design reviews and regular syncs, ensuring Sibel retained ownership of key technical decisions while meeting design requirements.

Audio & Haptics Software Engineer

Cupertino, CA

APPLE

March 2022 – September 2023

- Optimized Apple's CoreHaptics framework, achieving 20% less CPU usage across haptic and audio playback scenarios.
- Fixed a fundamental sound wave interpolation bug in Apple's haptic synthesizer, eliminating Taptic Engine motion glitches during sustained haptic playback.
- Improved iOS 16 haptic keyboard quality by validating behavior across all audio routes (wireless/wired headphones, CarPlay, AirPlay, HDMI) and expanded CI/CD test coverage for the CoreHaptics framework.

Firmware Engineer

Chicago, IL and Remote

SIBEL HEALTH

June 2020 – March 2022

- Led firmware development for the Maxim Integrated Health Patch Platform MAXREFDES282, a reference design showcasing the MAX816178 PPG, ECG, and BIOZ all-in-one AFE.
- Built a fleet management system to verify quality of medical sensors manufactured abroad, reducing defective shipments by 21%. Developed diagnostic firmware routines for PCB integrity validation and life-cycle stress testing.
- Implemented low-level drivers for sensing components (digital mics, multi-lead ECG AFE) and critical peripherals (NAND flash, wireless-charging PMICs).

Interests

Backpacking and outdoor living, The Sopranos, Seafood Foraging, and electronic music production